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Virtual Choreographies of Learning: AI-Driven Discovery in K–12 Online American Schools

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Resumo

As *Connections Academy*, apoiadas pela Pearson Online and Blended Learning, oferecem escolas totalmente online do Jardim-de-Infância ao 12.º ano em 32 estados dos EUA, servindo mais de 100 000 alunos, muitos dos quais transitaram do ensino presencial devido a problemas anteriores. Relatórios nacionais indicam piores resultados médios em exames padronizados para alunos de escolas online a tempo inteiro, embora estas escolas matriculem mais estudantes em risco, dificultando comparações diretas. Apesar da escala, existe pouca investigação sobre o que os alunos efectivamente fazem nestes ambientes e como os seus padrões de estudo e interação com docentes/colegas se relacionam com o progresso. Esta dissertação analisa dados das 32 escolas usando o enquadramento de *coreografias virtuais* do INESC TEC para identificar sequências recorrentes de ações de aprendizagem online e a sua ligação ao sucesso. Para tal, explora técnicas de IA como *k*-means, t-SNE/UMAP e modelos supervisionados (árvores de decisão e *random forests*) para descobrir *clusters* e trajetórias comportamentais menos óbvias e enriquecer a compreensão da dinâmica de aprendizagem em escolas K–12 totalmente online.

Abstract

Connections Academy schools, supported by Pearson Online and Blended Learning, provide fully online K–12 education across 32 U.S. states and currently serve over 100,000 students. Many families choose these schools because their children faced specific problems in prior in-person settings. National reports indicate that students enrolled in full-time online schools often perform worse on standardized exams than peers in traditional public schools; however, fully online schools enroll a higher proportion of at-risk students, which complicates fair comparisons.

Despite the scale of this model, there is limited research on what actually happens in these learning environments: Which patterns of study do students follow? How do they engage with teachers and peers? And how are these patterns connected to learning progress? This dissertation addresses these questions by analyzing data from the 32 Connections Academy schools using the *virtual choreographies* framework developed at INESC TEC. The goal is to identify recurring sequences of online learning actions and examine their relevance for student success.

To that end, the work explores Artificial Intelligence techniques to detect and characterize behavior patterns and trajectories, including unsupervised methods such as k -means and dimensionality-reduction techniques (t-SNE/UMAP), alongside supervised models such as decision trees and random forests. These approaches are expected to surface non-obvious configurations of behavior and to enrich our understanding of student learning dynamics in fully online K–12 schools.

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List of Acronyms

AI	Artificial Intelligence
ARI	Adjusted Rand Index
EDM	Educational Data Mining
EPM	Educational Process Mining
FEUP	Faculdade de Engenharia da Universidade do Porto
IEP	Individualized Education Program
K-12	Kindergarten to 12th Grade
LA	Learning Analytics
LMS	Learning Management System
ML	Machine Learning
NEPC	National Education Policy Center
RQ	Research Question
SHAP	SHapley Additive exPlanations
t-SNE	t-Distributed Stochastic Neighbor Embedding
UMAP	Uniform Manifold Approximation and Projection
URL	Uniform Resource Locator

Chapter 1

Thesis Manifest

1.1 Context

Connections Academy, supported by Pearson Online and Blended Learning, operates fully online public schools serving K–12 students across 32 U.S. states and enrolling over 100,000 learners. Independent national reviews repeatedly find that students in full-time virtual schools, on average, underperform peers in traditional brick-and-mortar public schools on standardized measures [10]. However, the literature suggests these schools often serve a distinct population of “at-risk” students who have opted out of traditional schooling due to medical, social, or academic challenges [2]. These complex demographics make it crucial to understand what students actually *do* in virtual classrooms—beyond simple attendance—and how their behavioral routines relate to learning progress.

1.2 Motivation

Despite the scale of full-time online schooling, there is limited empirical detail about students’ fine-grained learning activities—how they navigate content, participate in live sessions, and interact with teachers and peers—and how these behaviors connect to outcomes [8]. Addressing this evidence gap can inform practice in large online school networks and help target support to the students who need it most, particularly those whose engagement patterns signal early risk of failure.

1.3 Problem

This dissertation investigates *virtual choreographies*—recurring sequences of online learning actions—as a lens to characterize K–12 students’ behavior at scale and to examine how such patterns are associated with academic success. Concretely, the aim is to identify, describe, and relate behavior patterns observed in activity logs to indicators of performance in fully online schools, utilizing the platform-independent framework proposed by Cassola et al. [6].

1.4 Research Questions

RQ1 Identification. Which *virtual choreographies* (recurring sequences of online learning actions) can be reliably discovered in Connections Academy data using unsupervised clustering, and how stable are they over time?

RQ2 Robustness across contexts. To what extent do the discovered choreographies replicate across grades, subjects, schools, and cohorts (i.e., are they consistent behavioral habits rather than school- or cohort-specific artifacts)?

RQ3 Association with outcomes. How are choreography *attributes*—such as temporal regularity, balance between synchronous and asynchronous participation, and frequency of feedback—associated with indicators of student success (e.g., progress, grades), when controlling for student background?

RQ4 Predictive utility. Do choreography-based representations improve early prediction of student success over standard activity metrics, and how early in the semester can such signals be detected?

RQ5 Actionability and Visualization. Which components within the choreographies (specific actions, transitions, or rhythms) most contribute to outcomes, and how can these be visualized to support teacher decision-making?

1.5 Hypotheses

Definition (Virtual Choreography). A *virtual choreography* is the structured set of actions carried out among agents (e.g., student, teacher, platform tools) that unfolds over time and within a specific learning space, abstracting specific platform clicks into semantic behaviors [6].

H1 — Regularity. Choreographies exhibiting stable temporal rhythms (e.g., consistent study times and revisits across the week) are positively associated with academic success.

H2 — Balanced participation. Choreographies that balance synchronous participation (live sessions) and asynchronous work (content study, assignment submission) relate to better outcomes than predominantly one-sided patterns.

H_Robustness — Context Stability (Addressing RQ2). We hypothesize that choreographies associated with highly structured subjects (e.g., Mathematics) will exhibit higher stability across cohorts than those in open-ended or creative subjects, and that core behavioral habits will remain consistent across adjacent grade levels.

H3 — Feedback-centric interaction. Choreographies with frequent teacher–student feedback exchanges (asking questions, reading feedback, revising submissions) are positively associated with success.

H4 — Purposeful navigation. Choreographies characterized by efficient, task-aligned navigation (timely access to resources, limited detours) are associated with stronger progress than fragmented, sporadic engagement.

H5 — Risk signals (inverse). Choreographies marked by long inactivity gaps, last-minute bursts, or minimal interaction with teachers/peers correlate with weaker outcomes.

Chapter 2

Literature Review

2.1 Overview

This chapter describes the methodology used to identify and select the scientific literature relevant to this dissertation. It explains the type of literature review adopted, the search strategy, the inclusion and exclusion criteria, and the process followed to screen and select the final set of papers that compose the document corpus. In line with the thesis manifest, the review focuses on research about K–12 online learning and full-time virtual schools in the United States, with particular emphasis on their effectiveness, challenges and implications for students, teachers and school systems. This corpus provides the conceptual and empirical background needed to frame the later analysis of virtual choreographies of learning in Connections Academy schools.

2.2 Type of review

Given the objectives of this dissertation, a structured literature review was conducted, following principles of a systematic literature review while allowing some flexibility that is closer to a narrative synthesis. The goal of the review is to identify, select and document existing research on K–12 online learning and virtual schools, so as to support the definition of the dissertation research questions and to provide a solid basis for the later State of the Art chapter and for the empirical analysis of student activity data. The corpus prioritizes contemporary systematic reviews and empirical studies that reflect the current state of K–12 online learning technologies. Instead of early foundational work, the review relies on recent comprehensive analyses, such as the systematic reviews by Martin et al. [9] and Johnson et al. [8]. It also incorporates critical studies on student outcomes, engagement and school-level implementation of K–12 online programmes, specifically those addressing at-risk populations as explored by Beck [2, 3] and Toppin and Toppin [12], as well as earlier warnings about the maturity of the field [1].

2.3 Search strategy

The literature search was carried out in 2025, in the months leading up to this deliverable. The primary search engine used was Google Scholar. While discipline-specific databases like ERIC or Scopus are valuable, Google Scholar was selected as the primary source for its broader indexing of interdisciplinary research, capturing relevant studies at the intersection of Computer Science (Educational Data Mining) and Education that might otherwise be fragmented across distinct databases.

In addition, two discovery tools, Litmaps and ResearchRabbit, were used to expand the initial set of articles through citation-based exploration and visualisation of related work. The search targeted peer-reviewed journal articles, conference papers and book chapters written in English. Given the rapid evolution of educational technology and the impact of the COVID-19 pandemic, the search prioritized studies published between **2015 and 2025**. This timeframe ensures that the analyzed studies reflect the current technological infrastructure (modern LMS and learning analytics capabilities) and the post-pandemic reality of K–12 online education.

The main search string used in Google Scholar was iteratively refined, but was based on combinations of the following keywords:

- **Core topic terms:** “K-12 online learning”, “virtual schools”, “virtual schooling”, “online teaching”, “online school”, “distance education”.
- **Outcome / focus terms:** “effectiveness”, “student achievement”, “engagement”, “school choice”, “best practices”, “teacher training”, “at-risk students”.
- **Context terms:** “K-12”, “secondary education”, “United States”, “high school”.

An example of a typical query used in Google Scholar is:

“K-12 online learning” OR “virtual schools” OR “virtual schooling” AND (effectiveness OR achievement OR engagement OR “distance education”)

The initial Google Scholar searches generated a broad set of potentially relevant results. These references were exported and then imported into Litmaps and ResearchRabbit. From this initial seed set, both tools were used to identify additional papers that frequently cite or are cited by the seed articles, visualise clusters of research, and surface key works. Through this process, an initial pool of approximately 150 records was assembled before applying the inclusion and exclusion criteria described below.

2.4 Inclusion and exclusion criteria

To ensure the relevance and quality of the selected literature, explicit inclusion and exclusion criteria were defined and applied during the screening process.

2.4.1 Inclusion criteria

A study was included in the corpus if it met all of the following criteria:

- **IC1** – The study focuses on K–12 online learning, virtual schools or virtual schooling (fully online or primarily online programmes).
- **IC2** – The study is peer-reviewed (journal article, conference paper or book chapter).
- **IC3** – The study is written in English.
- **IC4** – The study was published between **2015 and 2025**.
- **IC5** – The study addresses at least one of the following aspects: effectiveness or outcomes of virtual schooling, teaching practices, student engagement, or system-level issues such as school choice and policy.

2.4.2 Exclusion criteria

Studies were excluded if any of the following applied:

- **EC1** – The main focus is higher education, adult education or corporate training.
- **EC2** – The publication type is a thesis, dissertation, report, blog post, or other non-peer-reviewed document.
- **EC3** – The study is not available in full text.
- **EC4** – The study deals with general educational technology or blended learning without a clear focus on fully or primarily online K–12 programmes.
- **EC5** – The paper is purely theoretical or opinion-based without sufficient connection to K–12 online or virtual schooling.

2.5 Screening and selection process

The selection process followed three main stages: (i) title and abstract screening, (ii) full-text assessment, and (iii) citation-based expansion and refinement.

2.5.1 Stage 1: Title and abstract screening

After removing duplicates, approximately 150 unique records remained. Titles and abstracts were screened against the criteria. Studies clearly focusing on higher education or non-online contexts were removed, leaving approximately 60 papers.

2.5.2 Stage 2: Full-text assessment

The full text of the remaining studies was examined. Papers that mentioned online learning only superficially were excluded. This stage reduced the set to a core group of studies providing substantial evidence, including recent systematic reviews [8, 9] and research on at-risk students [3, 4]. Around 20 papers remained.

2.5.3 Stage 3: Citation-based refinement

The core set was used for backward and forward snowballing using Litmaps. After applying the same criteria to these additional articles, the final corpus consisted of **12 highly relevant papers**.

2.6 Summary of Selected Studies

To provide a clear overview of the state-of-the-art, Table 2.1 summarizes the key papers selected for the final corpus.

Author (Year)	Type	Focus	Key Insight
Martin et al. (2020)	Review	Effectiveness	Effectiveness depends on design/support, not just the medium.
Johnson et al. (2022)	Review	Teaching Practices	Importance of teacher facilitation in online settings.
Beck (2023, 2024)	Empirical	At-Risk Students	At-risk students perform better with strong “advocate” support.
Curtis & Werth (2015)	Empirical	Engagement	Transactional distance leads to isolation; needs interaction.
Molnar et al. (2023)	Report	Policy	Virtual schools often lag in graduation rates compared to traditional.
Toppin & Toppin (2016)	Analysis	Demographics	Virtual schools attract students with prior academic/social issues.

Table 2.1: Summary of the core literature corpus.

2.7 Effectiveness of K-12 Online Learning

A central theme in the selected literature is the comparative effectiveness of full-time virtual schools versus traditional brick-and-mortar settings. Large-scale reports consistently highlight significant performance gaps [10], noting lower graduation rates. However, recent reviews [8, 9] caution against binary comparisons, suggesting effectiveness is dependent on instructional design and support. Johnson et al. [8] note that while average performance may be lower, online learning provides essential opportunities for credit recovery and advanced coursework.

2.8 Supporting At-Risk Populations

A critical finding is that the demographic profile of virtual schools differs significantly from traditional ones. Toppin and Toppin [12] and Beck [2] observe that virtual schools frequently attract

“at-risk” students—those with prior bullying, medical issues, or academic failure. Success for these learners relies heavily on support beyond the screen. Beck’s research [3, 4] highlights the pivotal role of the “on-site facilitator” (typically a parent). Beck and Levine [4] found that parental engagement was a stronger predictor of success than many course-level variables during the pandemic.

2.9 Engagement and Interaction Patterns

Student engagement is cited as the primary predictor of retention. Curtis and Werth [7] identify strategies to foster engagement, noting that “transactional distance” leads to isolation. However, a methodological gap exists: most studies rely on surveys. There is limited research utilizing fine-grained log data to understand *temporal patterns* of engagement. This gap underscores the need for the data-driven approach proposed in this dissertation.

2.10 Limitations and threats to validity

The review is subject to limitations. First, Google Scholar does not index all databases, potentially missing some studies. Second, the search was limited to English publications. Third, the focus on U.S. virtual schools excludes international contexts. Finally, subjective judgement was involved in screening. These limitations will be considered in the State of the Art analysis.

Chapter 3

State of the Art

3.1 Overview

While the previous chapter focused on the educational context of K–12 online learning and the needs of at-risk populations, this chapter reviews the technological landscape. Specifically, it addresses the field of Educational Data Mining (EDM) and the computational methods available for discovering patterns in student behavior. The chapter defines the scope of EDM, contrasts traditional Educational Process Mining with the proposed *Virtual Choreographies* framework, and reviews the unsupervised machine learning algorithms selected for this study.

3.2 Educational Data Mining

Educational Data Mining (EDM) is defined as the area of scientific inquiry centered on the development of methods for making discoveries within the unique kinds of data that come from educational settings [11]. In their comprehensive survey, Romero and Ventura [11] distinguish EDM from Learning Analytics (LA):

- **Learning Analytics (LA):** Often focuses on human-led decision making (e.g., visual dashboards for teachers) and relies on statistics.
- **Educational Data Mining (EDM):** Emphasizes automated discovery and the development of algorithms to find hidden patterns without human intervention.

In the context of K–12 online schools, EDM is typically applied to three main tasks [11]:

1. **Prediction:** Using historical data (grades, login frequency) to forecast student outcomes or dropout risk.
2. **Structure Discovery:** Finding underlying structures in data, such as grouping students with similar learning strategies.
3. **Relationship Mining:** Identifying relationships between variables.

This dissertation focuses primarily on *Structure Discovery*, using unsupervised learning to find patterns not immediately obvious to teachers.

3.3 Modeling Student Behavior

To analyze student engagement beyond simple metrics, researchers use techniques to model the *sequence* of student actions.

3.3.1 Educational Process Mining (EPM)

A dominant approach is Educational Process Mining (EPM). As detailed by Bogarín et al. [5], EPM applies process mining techniques to educational data, treating learning as a series of events (e.g., Quiz Start $\rightarrow$ Quiz End). While effective for structured tasks, Bogarín et al. [5] note that EPM faces challenges in flexible environments. In full-time virtual schools, EPM often results in “spaghetti models”—complex, tangled diagrams that are difficult to interpret. **This lack of readability makes such models unsuitable for teachers who need clear, actionable insights (addressing RQ5), necessitating a higher-level abstraction like Virtual Choreographies.**

3.3.2 The Virtual Choreographies Framework

To address the limitations of rigid process maps, this dissertation adopts the concept of **Virtual Choreographies**. Cassola et al. [6] define a *Virtual Choreography* as a platform-independent representation of actions, interactions, and events that unfold over time. Unlike raw clickstreams, a Virtual Choreography maps technical events into **semantic actions** (e.g., mapping URL : /math/quiz/1 to Assessment). This abstraction allows for:

- **Platform Independence:** The analysis focuses on the behavior, not the system.
- **Routine Discovery:** It enables the identification of clusters of students who share similar daily routines (e.g., “Late Night Crammers” vs. “Steady Workers”) regardless of the specific course.

3.4 Unsupervised Learning Techniques

The discovery of these choreographies relies on unsupervised machine learning algorithms.

3.4.1 Clustering Algorithms

Clustering groups objects so that objects in the same group are more similar to each other than to those in other groups. In EDM, the most widely used algorithm is **k-Means**, due to its computational efficiency and interpretability [11]. However, k-Means requires the number of clusters (k) to be specified, necessitating the use of validation metrics such as the Elbow Method or Silhouette Score.

3.4.2 Dimensionality Reduction

Student behavioral data is often high-dimensional. To visualize clusters effectively, dimensionality reduction is required. **t-SNE** and **UMAP** are state-of-the-art non-linear techniques used to project high-dimensional data into 2D space. These visualizations allow researchers to “see” the separation between different student groups and validate whether the identified choreographies represent distinct behavioral patterns.

3.5 Summary

Current research highlights the need for better support systems for at-risk students. Technologically, while EPM offers tools to analyze sequences, it can be overly rigid for online schooling. By adapting the *Virtual Choreographies* framework [6] and combining it with standard EDM clustering techniques [11], this thesis aims to discover interpretable patterns of student behavior that can inform the educational strategies discussed in the literature review.

Chapter 4

Methodology

4.1 Overview

This chapter describes the methodological process used to analyze, characterize, and predict the academic success of students in fully online schooling. The pipeline comprises this stages: data extraction and preparation, pattern discovery through k-means clustering and UMAP visualization, association of the discovered patterns with academic outcomes, and prediction of student success using Random Forests.

4.2 Synthetic Data as a Methodological Choice

The data used in this dissertation is synthetic. The real datasets were expected first from Pearson and subsequently from Stride Inc., but neither could be delivered in time. The synthetic dataset had originally been created as a working substrate to allow development to proceed while the real data was pending; once it became clear the real data would not arrive, continuing with the synthetic data was the only viable path. This choice carries a decisive methodological advantage. Because the dataset was generated by the author, the true behavioral archetype of each student is known by construction. This ground truth, which is never exposed to the clustering algorithm, makes it possible to measure whether the discovered clusters actually recover the underlying behaviors, using metrics such as the Adjusted Rand Index (ARI). With real, anonymized data, these true labels would be unknown, and such a quantitative validation of the clustering would not be possible. A second advantage is openness. As the data does not belong to real students or institutions and was created entirely by the author, it can be used and published without privacy or licensing constraints, supporting full reproducibility of the pipeline. This approach has an explicit limitation. Synthetic results cannot be validated in the same way as real data, and they do not constitute empirical claims about real Connections Academy students. What they validate is the methodology, the pipeline and analytical methods developed to carry out the study. The generator's parameters were designed through informed intuition rather than calibrated against measured empirical distributions. The five archetypes were specified so that each would be behaviorally plausible and

clearly distinct from the others, drawing on the patterns of engagement and disengagement discussed in the literature reviewed earlier drawing on the patterns of engagement and disengagement discussed in the literature reviewed earlier [3, 7]. They are not intended to reproduce exact empirical frequencies from real online schools, but to provide coherent, recognizable behavioral profiles against which the discovery pipeline can be developed and validated.

4.3 Synthetic Data Generation

The synthetic generator builds an academic year of activity for thousands of students, each assigned to one of five behavioral archetypes that describe how they study. The Steady Worker is a consistent, engaged student, focused on content study, assignments, and assessments. The Balanced Engager is the most social and interactive profile, driven by live sessions and discussion forums. The Night Crammer studies intensively, but concentrates activity late at night or in the early morning. The Minimal Browser mostly navigates pages, checks the dashboard, and looks up grades, with little actual studying. Finally, the Disengaged Ghost is a near-absent student, showing minimal activity of any kind. Students are sampled from these archetypes with embedded correlations. At-risk students are more likely to be assigned to the lower-engagement archetypes, reflecting the weaker outcomes associated with these profiles. The dataset reproduces a realistic institutional structure: 32 schools across 32 U.S. states, spanning 13 grade levels from Kindergarten to Grade 12. Activity is generated as event-level logs, one student at a time. Each event records an action type, drawn according to the student's archetype, along with the time of day and, where applicable, the session duration. A fixed random seed (42) is used throughout the generation process to ensure full reproducibility.

4.4 Feature Engineering

The unit of analysis is the student-week: one row per student per calendar week. This choice provides enough data points for the clustering algorithm to find patterns, while letting each student be represented as a sequence of weeks across the year. Two alternatives were considered, the student-day and the student-year, but the former would generate an excessive volume of sparse data, and the latter would not retain enough observations per student. The student-week was the most balanced option. Weekly features follow a two-block design. The WHAT block captures the proportion of each semantic action type performed during the week (for example, how much content study, assessment, or forum activity). The WHEN block independently captures the proportion of activity falling in each of four time-of-day bins: Morning, Afternoon, Evening, and Night. The two blocks are kept separate rather than crossed. In an early attempt, action and time-of-day were combined into a single set of features, which caused the time dimension to dominate k-means: the resulting clusters separated students almost entirely by period of day rather than by behavior. Decoupling the two blocks resolved this and let the clustering focus on what students do, with time of day as a secondary signal. Login and Logout actions are deliberately excluded

from the WHAT block. They are universal, every student performs them, and therefore carry no discriminative value; they merely mark session boundaries. Finally, aggregate metadata (number of sessions, active days, total actions) is computed but kept outside the clustering feature space, so that students are grouped by the shape of their behavior rather than by sheer activity volume.

4.5 Pattern Discovery (Clustering)

At this stage, the goal of clustering is to discover the shape of behavior, for which k-means was chosen as a clustering algorithm widely used in educational data mining [11]. Because k-means groups data by computing mathematical distances between points, raw action counts would let differences in scale dominate: the algorithm would simply separate students who click a lot from students who click little, failing to capture the actual behavioral patterns. To prevent this, each weekly vector is L1-normalized into proportions. After normalization, two students with the same behavioral mix, for instance, both spending half their activity on content study and half on forums, are treated as similar even if one performed many more actions than the other. Clustering therefore focuses on the composition of behavior rather than its volume. Standardization is then applied so that each feature contributes comparably to the distance computation, preventing frequent actions from overwhelming rarer but discriminative ones. This step, however, required care. Some action types are very rare, almost no student performs them, so their columns have a near-zero standard deviation. Standardizing divides by the standard deviation, so these columns would trigger division-by-zero errors that corrupt the model. The `safe_standardize` function avoids this by detecting near-constant columns (standard deviation below a small threshold) and neutralizing them to zero rather than dividing by them, so they simply carry no weight in the clustering. The number of choreographies, k , was selected by evaluating solutions across a range of $k = 2$ to 10, computing both the inertia (Elbow method) and the Silhouette score. These automatic metrics did not, on their own, give a clean answer. The Silhouette score favored sharp mathematical partitions, but the UMAP projection showed that behaviors merged continuously rather than forming isolated islands; relying on the metric alone would have split genuine profiles in half. Both $k = 4$ and $k = 5$ were therefore examined directly. Validation against the ground-truth archetypes showed that the two solutions recovered the original behaviors with essentially the same accuracy ($\text{ARI} \approx 0.53$). The deciding factor was interpretability: at $k = 5$, two of the clusters were both low-engagement profiles, separated only by time of day, which made them redundant. The final choice of $k = 4$ retains the same fidelity to the underlying structure while producing distinct, non-redundant choreographies. The final clustering was produced by fitting k-means with $k = 4$, ten initializations (`n_init = 10`), and a fixed random seed (42) to guarantee reproducible results. For visual diagnosis, the resulting clusters were projected into two dimensions using Uniform Manifold Approximation and Projection (UMAP), with `n_neighbors = 15` and `min_dist = 0.1`. It is important to stress that UMAP was used strictly for human inspection: it allowed visual verification of whether the discovered groups separated cleanly, but had no influence whatsoever on the

cluster assignments, which were determined solely by k-means. Removing the projection would leave the clusters unchanged.

4.6 Evaluation Design

RQ1 (Identification) was evaluated by comparing the algorithm's cluster assignments against the generator's ground truth. A contingency table (cluster $\times$ archetype) was built, and the purity of each cluster was computed, identifying the dominant student profile within each discovered group. To assess this correspondence quantitatively, two agreement metrics were applied: the Adjusted Rand Index (ARI) and Normalized Mutual Information (NMI). These metrics quantify the extent to which the algorithm recovered the behavioral structure originally embedded in the synthetic dataset. RQ2 (Robustness) was assessed along two dimensions. Temporal stability was measured over the full academic year using week-over-week ARI and the percentage of students retained in the same cluster from one week to the next. A transition matrix was built to map how students moved between choreographies; the concentration of values along the main diagonal indicates the degree to which students maintained consistent learning routines rather than shifting abruptly between profiles. Cross-context stability examined whether the discovered patterns were universal rather than demographic artefacts. The percentage distribution of each choreography was evaluated across grade levels and states, checking that the standard deviation across contexts remained small. Finally, to assess whether the shape of behavior changed with schooling level, the cosine similarity of each profile's behavioral signature was computed across grade bands (K-5 versus 6-12), measuring how closely the underlying structure of these profiles is preserved. RQ3 (Association with Outcomes) was assessed by first identifying each student's dominant choreography, the profile in which they spent most of their weeks across the year. With students grouped by dominant profile, a one-way ANOVA was applied to their final grades to test whether the performance differences between profiles were statistically significant, and the eta-squared statistic was used to measure the magnitude of the effect. Finally, to make the comparison fairer, these results were stratified by the student's initial at-risk status. This made it possible to examine to what extent final grades reflected the learning routines themselves rather than difficulties the student already carried beforehand. RQ4 (Predictive Utility) was evaluated by training a Random Forest classifier to predict whether a student would pass, using features drawn only from the first 4 and 8 weeks of the year. This strict temporal cutoff prevents data leakage: it ensures the model cannot "peek" at end-of-year behavior to anticipate the outcome. Two models were compared using the ROC-AUC metric: a baseline using only activity-volume features, and an experimental model adding the student's dominant choreography. ROC-AUC was chosen over accuracy because the classes are imbalanced (most students pass), which would make accuracy misleading. RQ5 (Actionability) was addressed by analyzing the trained prediction model to identify which actions carried the most weight in the final outcome, measuring feature importance in two ways: Gini importance and permutation importance. Two approaches were used because Gini importance has a well-known bias: it tends to overvalue features with many distinct values, such as navigation clicks, which can

be misleading. Permutation importance is more reliable, as it measures the actual drop in model performance when a feature is shuffled. Once the most important behaviors were identified, the direction of effect was examined, checking whether a high proportion of a given action was associated with passing or failing, which yields the practical, actionable insights about which behaviors to encourage. Mean absolute SHAP values (via a TreeExplainer) were also considered as a means of corroborating these rankings.

4.7 Reproducibility and Implementation Details

The pipeline can be reproduced in full, from start to finish. The guiding rule was to fix the random seed (42) at every stage of the project, from the generation of the synthetic students to the train/test split in the Random Forest. This forces the computation to make exactly the same decisions on any machine, guaranteeing identical results across runs. The analysis is organized as a sequence of Python scripts that must be executed in a defined order: synthetic data generation, feature engineering, clustering, validation, stability analysis, outcome association, early prediction, and interpretation, documented in the repository. The entire work was made available in a public repository, relying only on standard Python libraries (pandas, numpy, scikit-learn, umap-learn, matplotlib, seaborn, scipy).

Chapter 5

Results

5.1 Overview

5.2 RQ1: Identification of Choreographies

5.3 RQ2: Robustness (Temporal and Cross-Context)

5.4 RQ3: Association with Outcomes

5.5 RQ4: Predictive Utility

5.6 RQ5: Actionability and Interpretation

Chapter 6

Discussion

6.1 Interpretation of Findings

6.2 Revisiting the Hypotheses (H1–H5)

6.3 Relation to Prior Work

6.4 Limitations

6.5 Threats to Validity

Chapter 7

Conclusion

7.1 Summary of Contributions

7.2 Future Work

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